

MEMORY — COMPLETE OPEN-BOOK MASTER NOTE (PSY 101)

1. WHAT IS MEMORY?

Memory is the process of encoding, storing, and retrieving information.

We remember some experiences and forget others due to attention, meaning, interference, and organization.

2. BOTTOM-UP & TOP-DOWN INFLUENCES

Bottom-Up: raw perceptual input (what we see/hear).

Top-Down: prior knowledge, expectations, experience influencing memory.

3. THREE-STAGE MEMORY PROCESS

Encoding: transforming sensory input into memory.

Storage: maintaining information over time.

Retrieval: accessing stored information.

4. THE THREE-STAGE MODEL

Sensory Memory → Short-Term Memory → Long-Term Memory.

Rehearsal moves STM to LTM.

5. SENSORY MEMORY

Brief storage of sensory impressions (visual, sound, touch, smell, taste).

Most details fade quickly but helps recognize voices/colors.

6. DIGIT SPAN

Average STM capacity: 7 ± 2 items.

Examples:

Span 4: 6194

Span 5: 37852

Span 6: 965283

Span 7: 4269851

Span 8: 81637249

7. SHORT-TERM MEMORY (STM)

Limited capacity: 7 ± 2 chunks.

Chunking uses prior knowledge.

Can miscode based on associations.

8. LONG-TERM MEMORY

Implicit Memory (Unconscious): skills, habits → “knowing how”

Explicit Memory (Conscious): facts, events → “knowing that”

Examples:

Implicit: riding a bike.

Explicit: remembering a birthday.

9. ORGANIZATION & CHUNKING

Letters: GDCEIAHBF vs ABCDEFGHI (better organized).

Words grouped by meaning = easier recall.

10. PRIMACY & RECENCY EFFECTS

Primacy = items at beginning → LTM.

Recency = items at end → STM.

Explains why middle items often forgotten.

11. LEVELS OF PROCESSING

Shallow Encoding (Appearance): poor memory.

Deep Encoding (Meaning): strong memory.

Example: living vs non-living classification improves memory.

12. EBBINGHAUS' CURVE OF FORGETTING

Forgetting is rapid initially, then slows.

Immediate recall highest; drops within hours.

13. INTERFERENCE

Proactive: old info blocks new learning.

Retroactive: new info blocks old learning.

14. FLASHBULB MEMORIES

Emotionally charged memories.

Features: surprise, arousal, context, source.

Accurate but still vulnerable to distortion.

People are very confident in them.

15. IMPROVING MEMORY (PSYCHOLOGY-BASED)

Study repeatedly.

Make material meaningful.

Activate retrieval cues.

Use mnemonics.

Minimize interference.

Sleep enough.

Self-test.

Exercise improves blood flow & brain connections.

Elaborative rehearsal = move STM to LTM.

Self-reference effect = link info to yourself.

Reduce distractions.

Mental activity, social interaction, organization, healthy diet.